Client Data

Audit, Copay, Consulting Clients (~300 total, past and present)

Client data files (Rx claims, eligibility, specialty drug lists, MAC lists, etc.)

Copy file / Send

FTP Server (client FTP preferred, else PillarRx FTP on files.com)

Secure Email

Receive

External Data

Wolters Kluwer Medi-Span

fdb First Databank

NCPDP

NPPES National Plan & Provider Enumeration System

CMS.gov Centers for Medicare & Medicaid Services

Drug reference files

Pharmacy reference files

Provider reference files

CMS data and reference files:

LISHIST (Low-Income Subsidy History)
MONMEMD (Monthly Membership Detail)
FIR (Financial Information Reporting)
PDE (Prescription Drug Event)
LEIE (OIG List of Excluded Individuals and Entities)

Blob Storage

Copy file (manual)

PillarRx ETL Team

ADF Copy file

Azure Blob Storage

HITL

Clean schema translation

Agnostic Ingestion

Schema registry

Extraction

Data Warehouse / Lake

ADF ETL

Business App Data

Schema: dw_acq (Acquisition)
Purpose: Collect raw data and track version history on data changes where applicable. Support prototyping efforts for dw_pres layer.
Structure: Independent tables (no relationships)
Organization: Separate table for each source data format
Version history: Type-2 is the default, but may be type-1 for some tables if source tracks version history, or is insert only, or no version history is needed/desired.

ETL ADF

Schema: dw_int (Integration)
Purpose: Limited use (as needed) when ETL logic to load a dw_pres object is sufficiently complex to warrant additional steps and persisting of data
Structure: Clusters of related tables, may be 3NF or denormalized
Organization: As needed
Version history: Type-1 is the default, type-2 where version history is needed/desired.

ETL ADF ETL ADF

Schema: dw_pres (Presentation)
Purpose: Present conformed, consistent, integrated data to downstream applications, reports, and analytics
Structure: Dimensional
Organization: Conformed by subject area (source data formats harmonized into common target model)
Version history: Type-1 is the default, type-2 where version history is needed/desired.

DB views or materialized views

Schema: dw_lake (Lake)
Purpose: Make raw data available to power users who need to access source format-specific attributes not contained in the dw_pres layer, typically for rare / one-off analysis.
Structure: Same as dw_acq
Organization: Same as dw_acq
Version history: Same as dw_acq

Query

Azure SQL DB (PaaS)

DB power users (Analytics, Ops, ETL, etc.)

DB Name: sqldb-dw-prd-eastus2-001

Business Apps

ADF Copy Data (daily push per client; will require changes from current state)

ETL ADF (Copay enrollments / exceptions, drug lists, etc.)

MemberOps DB (no change from current state)

SQL Server instance: SourceDataDB01

Copay Users — Read Rx claims, eligibility Create enrollments — Interact with app

PAX Web App (for copay assistance program)

Scheduled pull from Qlik

Copy Data (triggered on request by Accucast app; will require changes from current state)

ETL ADF (Repriced Rx claims, audit results, etc.)

Accucast DataStore Project DB 1-N (no change from current state)

Audit Users — Multiple audits being worked in parallel; each uses one Project DB — Read Rx claims, Write repriced Rx claims

Copy Data (triggered on request by Accucast app; will require changes from current state)

TricastBenchmarksDB (no change from current state)

Read Drug reference data

Auditing Web Apps: AccuCast, Insight, Pharmacast

Read/Write app config data (process runs, plan designs, lists, etc.)

Extract

ADF Extract files (monthly copay savings, accumulators, hyperstation, etc.) Copy File

Retrieve File

FTP (note: shown here for convenience; technically not in Azure cloud)

Audit, Copay, Consulting Clients

TricastMaster DB (no change from current state)

Azure SQL VM (IaaS)

SQL Server instance: AccucastDB

Microsoft Azure Cloud

BI / Reporting

Qlik Data File (QDF) Container

Read Qlik Data Files (QDFs), build in-memory data set

Qlik Sense BI tool (SaaS)

Self-service, In-memory analytics

Scheduled or On-Demand

Reports (monthly consulting reports, audit reports, etc.)

View report

Audit, Copay, Consulting Users

Send reports, custom analyses, etc.

Audit, Copay, Consulting Clients

AWS Amazon Web Services Cloud

Line of Business Extraction

Legend

- Data flow
- App-to-DW data flow
- Reporting / app flow
- ADF — Azure Data Factory (ETL tool)